

Monash University
FIT5147 Data Exploration and Visualisation
Semester 1, 2026

Programming Exercise 1: Tableau (5%)

Please carefully review all the requirements below to ensure you have a good understanding of what is required for your assessment.

- 1. Instructions & Brief**
 - 2. Assessment Resources**
 - 3. Assessment Criteria**
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1. Instructions & Brief

In this assignment, you are required to read in some data then explore and visualise it using Tableau Public or Tableau Desktop **only**. You will submit a report showing your findings and the visualisations you used. **It is an individual assignment and worth 5% of your total mark for FIT5147.**

Relevant learning outcomes for FIT5147:

1. Perform exploratory data analysis using a range of visualisation tools;
6. Implement interactive data visualisations using R and other tools. (*restricted to Tableau*)

Originality:

This is an individual assignment and must be original. Any sources you use to aid in the completion of the assignment must be acknowledged appropriately to ensure the academic integrity of your work. See the assessment page on Moodle for more information, including citing sources. If you have used **Generative AI** you must declare the tool(s) and prompt(s) used at the end of your report. This includes using any software that paraphrases, translates or rewrites your text.

THE DATA:

The data set used in this assignment is based on the [Atlas of Living Australia \(ALA\)](#) online resource, which is “a collaborative, digital, open infrastructure that pulls together Australian biodiversity data from multiple sources, making it accessible and reusable” (ALA, n.d.) and is regularly updated. The ALA relates far back in history, depending on the species and the source of the data.

We introduced this data source in the Week 1 Workshop. **For this assignment, we provide you with a subset of the ALA dataset that has been slightly modified.** This modified dataset can be found on Moodle in the Assessments section under the Programming Exercise 1 heading.

To enhance your understanding of the context and metadata, you can check the data source link. Using the various interactive tools for the data source of the full dataset may help enrich your visual analysis, for example the ALA has a map that allows you to explore the data spatially: <https://spatial.ala.org.au/>. If you discuss or replicate the visualisations or metadata provided by ALA, be sure to reference these correctly in your report¹.

For this PE1 assignment, the resulting data describes when and where the Swift Parrot (*Lathamus discolor*) has been seen. There are reports from far back in human history to the present year. For this assignment, we have extracted certain columns for your exploration, with the following descriptions from the ALA metadata provided below.

In order to complete this task, you may need to add your own columns derived from this subset only. No additional data is required. Wider research to provide context to help you understand your insights may be needed, but will be minimal. Whilst the full history of the Swift Parrot observations in this collection is provided in the data set for your interpretation, to complete the assignment tasks and questions you may choose to filter the time period to a more recent time period. If you do so, you must retain at least 2 decades of data and must explain and justify your choice of temporal subset in your report.

Table 1: Fields of the “ALA_S12026PE1” data set

Column	Description
dataResourceUid	A data resource (“dr”) ID consisting of a combination of categorical and numerical values.

¹ ALA does include a [citation guide](#), but you may have to adapt the formats they supply to those specified for this unit.

dataResourceName	The data resource that supplies the record. This is typically a CSV or DwCA file, or a web service.
basisOfRecord	What this is a record of e.g. specimen, human observation, fossil - http://rs.tdwg.org/dwc/terms/basisOfRecord
observationDate	The date of observation or specimen collection date. (Extracted from the EventDate column)
stateProvince	The state or territory matching the supplied coordinates - http://rs.tdwg.org/dwc/terms/stateProvince
recordID	Atlas unique ID for this record - http://rs.tdwg.org/dwc/terms/recordID
decimalLatitude	The decimal latitude associated with this record. If the original latitude was supplied in degrees, minutes, seconds, this value represents the converted value. - http://rs.tdwg.org/dwc/terms/decimalLatitude
decimalLongitude	The decimal longitude associated with this record. If the original longitude was supplied in degrees, minutes, seconds, this value represents the converted value. - http://rs.tdwg.org/dwc/terms/decimalLongitude
coordinateUncertaintyInMeters	A radius in metres defining the smallest circle the location could occur within - http://rs.tdwg.org/dwc/terms/coordinateUncertaintyInMeters
scientificName	The name the Atlas has matched this record to in the NSL - http://rs.tdwg.org/dwc/terms/scientificName
vernacularName	The common name the ALA has matched this record to in the NSL - http://rs.tdwg.org/dwc/terms/vernacularName

In this data file there are some **irregularities or errors**. One of the requirements of this assignment is for you to find (using data visualisation), describe and handle them.

References:

ALA. (n.d.). Atlas of Living Australia (ALA), Retrieved February 27, 2026, from <http://www.ala.org.au>

VISUAL ANALYSIS QUESTIONS TO BE ADDRESSED

Using the data provided and visual analytics, you will need to answer the following questions:

1A. When were these birds observed over time?

To answer this question, you should identify, measure and visualise the temporal trend in observations. This should include observations per *year* and *month*. Discuss what you observe in your visualisation(s).

1B. How have the number of observations from the different data resources in this collection changed over time?

To answer this question, you should identify, measure and visualise the data resources and any significant changes over time. Discuss what you observe in your visualisation(s).

1C. In which Australian state and territories were these birds mainly spotted?

To answer this question, you should visualise the observations using the `stateProvince` values. Discuss what you observe in your visualisation(s).

2. Based solely on the visualisations and findings for 1A-C, is it possible for you to determine which data resources best show seasonal migrations of these birds?

For this question, you need to discuss whether your visual analytics for 1A-C have enabled you to answer this question or not. Be sure to explain how you came to that conclusion. To answer this question we do not expect any new visualisations but a reflection on your prior observations.

ASSESSMENT TASK

The task has two components: **data exploration using Tableau**, and a **written report**.

Data Exploration using Tableau:

The steps you are expected to complete:

1. Load the dataset in Tableau Public/Desktop.
2. Use data visualisation *in Tableau* to **check for and find at least two** aforementioned irregularities in the dataset. Each type of irregularity may occur multiple times in the data. See important note below for some specific details.
3. Amend the data to **correct** these irregularities *using any tool of your choice* (e.g., Excel, Python, R, Tableau) and **justify** your choice of correction based on the irregularity.

4. Use *Tableau* to **create one or two visualisations per question** (not more than 2 per question) to **conduct your visual analysis** and answer questions 1A-C and 2. Remember to select appropriate visual variables to suit the data and your chosen visualisation.
5. Polish up your visualisations for presentation, e.g., add a suitable title, correctly label your axis, make sure labels and values are not truncated, include a legend. Ensure the font, font size and colour are suitable and legible for your report.
6. Write a report that **presents** and **describes** your data exploration process and the visual analysis it included. See below for details.

Your exploration must also be submitted as a Tableau workbook file (*.twb suffix). You do not need to submit it with the original or cleaned data (but the workbook is expected to be able to read from one of them in the same directory).

Important note for Step 2: indications of missing data like *NULL/null* **should not be regarded as irregularities** for this assignment. Preserved specimens (see *basisOfRecord*) in museum collections or alternatives may have inaccurate geographical locations but this should not be regarded as an irregularity for this assignment. If Tableau has any issues automatically recognising any date or time information, then this is not to be regarded as an irregularity for Step 2 but can be corrected by you in Tableau.

Written Report

Once you have finished your data exploration, write a report that contains the following information:

1. **Data loading, checking and cleaning** (i.e., Steps 1 to 3)
 - **For each of the errors or irregularities** that you have found, include a brief explanation (maximum of **one paragraph per irregularity**) and **an accompanying image** showing how you found them **using Tableau**, and **explaining & justifying** how you resolved them. The image must show a **relevant visualisation**, not just the data or a table.
2. **Data exploration and presentation** (i.e., Steps 4 and 5)
 - Include an explanation of what **insights** you have found out through the visual analysis to answer the questions. This should include:
 - Your **answer** to each question, based on your visualisation(s). Include **figures** for 1 or 2 relevant visualisations per question.
 - A **description** of your visualisation(s) and how they relate to the data and question (i.e., why it is an appropriate visualisation choice)
 - **Justification** of your visualisation(s) and choice of visual variables
 - Any **further insights, or issues** that you have identified from the data or visualisation(s) while answering the question.

The report should:

- Be submitted as a **PDF file**
- Be **no more than 5 pages** in length, including figures, with a minimum font size of 10 (title page, any table of contents, bibliography or declaration of AI usage are excluded from the page limit)
- Be **properly structured** with headings, subheadings, figure captions (in-text referencing of captions), page numbers, and references (where appropriate)
- Have **high quality images** of your visualisations with clearly readable and legible text/labels (presume that it is read as part of an A4 document with no zooming).
- You must use proper **academic referencing** for all reports in this unit. This should follow either the APA or IEEE structure as recommended by the Faculty. Use the [library's referencing guide](#) for support.
- Any use of tools with generative AI **must be declared at the end of your work**, including identifying the prompt(s) used and their purpose. This includes using such tools for writing, translating, coding, images or any other function.
- Not include any code snippets **except** for key Calculated Fields in Tableau.

2. Assessment Resources

- **ALA_S12026PE1.csv** (Available on Moodle)
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3. Assessment Criteria

The following outlines the criteria which you will be assessed against. The focus of the marker will be on what you have included in your report, but your submitted Tableau Workbook may be examined if there are any concerns with the academic integrity of your work.

- Demonstrated ability to check and clean data and read into Tableau [1%]
 - Demonstrated ability to appropriately visualise data for data exploration using Tableau [2%]
 - Demonstrated ability to see trends/patterns in data [1%]
 - Quality of report [1%]
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4. How to Submit

Once you have completed your work, take the following steps to submit your work.

1. Save your report as a .pdf file.

2. Name your file using the following structure **PE1_Surname_StudentID**
3. Save your Tableau workbook as a .twb file.
4. Compress the .twb workbook file into a .zip file so it can be submitted to Moodle. DO NOT include your report or any data in your zip file, only your Tableau workbook.
5. Name your zip file using the following structure **PE1_Surname_StudentID**
6. Click the **Add Submission** button on Moodle to submit and upload your report and workbook

5. Word Count & Late Penalty

The report must **not be more than 5 pages** of graded material including figures (min. font size 10). Up to 2 additional pages may be used if you wish, but restricted to:

- 1 page prior to the report as a title page with a table of content.
- 1 page after the report only for references and declarations.

1 mark (out of the total of 5) will be deducted if the report does not meet these requirements.

As per Monash policy: All late submissions will receive a penalty of 5% per day (0.25 marks per day out of a total of 5 marks) late inclusive, including on weekends. Work submitted more than seven days after the due date will not be marked.